

HOTAN, China

WMO: 518280

Lat: 37.133N

Lon: 79.933E

Elev: 4511

StdP: 12.45

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.6	14.4	-4.4	5.1	26.7	-1.1	6.1	29.5	10.0	31.0	9.4	30.0	3.1	250	0.408

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.9	95.7	63.7	93.2	63.1	90.7	62.5	68.1	87.4	66.5	86.1	65.2	84.8	5.4	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.7	97.3	77.6	59.8	90.7	75.4	58.2	85.8	74.7	35.3	87.5	33.9	86.6	32.7	84.9	75.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.3	10.6	9.4	8.6	100.6	6.1	2.1	4.2	102.1	0.6	103.4	-2.8	104.5	-7.3	106.0		
			7.4	71.0	5.7	2.4	3.3	72.7	0.0	74.2	-3.2	75.5	-7.3	77.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	57.4	25.9	35.8	51.7	63.9	71.4	77.2	80.1	78.2	70.7	58.9	44.2	30.0
	DBStd	19.76	7.33	7.52	7.87	7.34	6.07	5.36	5.41	5.27	5.13	6.22	7.27	6.34
	HDD50	2054	747	399	76	6	0	0	0	0	0	5	200	621
	HDD65	4496	1212	817	416	102	19	2	0	0	12	206	624	1086
	CDD50	4763	0	1	127	423	662	815	932	875	620	282	26	0
	CDD65	1730	0	0	2	68	216	366	467	411	182	18	0	0
	CDH74	16608	0	0	28	557	1818	3626	5160	4052	1260	107	0	0
	CDH80	6713	0	0	3	132	587	1507	2421	1712	344	7	0	0
Wind	WSAvg	4.6	3.6	4.1	4.7	5.2	5.5	5.7	5.1	4.9	4.6	4.2	4.0	3.7
Precipitation	PrecAvg	1.5	0.0	0.1	0.1	0.1	0.3	0.3	0.2	0.2	0.1	0.0	0.0	0.0
	PrecMax	4.4	0.4	0.7	0.7	0.9	1.7	1.3	1.0	0.9	0.9	0.6	0.3	0.7
	PrecMin	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	0.9	0.1	0.2	0.2	0.2	0.4	0.4	0.3	0.2	0.2	0.1	0.0	0.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.6	60.2	77.4	88.1	92.3	97.3	99.8	97.9	90.4	80.6	68.8	53.8
		MCWB	35.9	40.7	48.6	55.4	58.8	61.9	64.3	64.7	60.4	54.5	46.0	37.0
	2%	DB	44.4	54.4	72.5	83.7	88.8	93.5	96.7	94.2	86.8	77.1	63.4	47.0
		MCWB	31.6	38.0	46.5	53.4	57.2	61.2	64.5	64.0	59.7	53.2	43.6	33.8
	5%	DB	39.5	50.4	69.0	80.2	86.1	90.9	94.0	91.5	83.9	73.9	59.2	42.5
		MCWB	29.0	35.9	45.4	52.0	56.8	61.0	64.1	64.1	58.9	52.1	41.6	31.6
	10%	DB	35.7	46.9	65.0	76.7	83.3	88.2	91.1	88.7	81.2	70.4	55.6	38.9
		MCWB	27.1	34.2	43.7	50.6	56.1	60.6	63.7	63.6	58.3	50.2	39.9	29.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.9	41.7	49.5	56.6	62.0	66.7	71.2	70.5	64.9	56.4	46.9	37.5
		MCDB	51.9	56.9	73.8	84.0	85.4	87.9	89.6	88.9	83.8	76.1	66.9	52.1
	2%	WB	31.7	38.9	47.3	54.7	59.9	64.7	68.7	68.0	62.4	54.5	44.0	34.2
		MCDB	43.7	52.5	70.0	80.6	83.1	85.6	87.8	87.2	80.0	74.0	62.1	46.4
	5%	WB	29.3	36.7	45.7	53.1	58.5	63.2	67.0	66.4	60.9	52.9	42.0	31.8
		MCDB	39.0	49.0	67.2	76.5	80.8	84.1	86.6	85.6	78.8	71.6	58.1	41.4
	10%	WB	27.2	34.7	44.0	51.7	57.3	62.0	65.6	65.0	59.6	51.3	40.2	29.9
		MCDB	35.1	46.3	63.7	73.7	78.8	82.8	85.3	84.3	76.9	68.8	54.7	38.4
Mean Daily Temperature Range	5% DB	MDBR	14.0	15.1	17.8	18.9	19.0	18.1	17.9	17.3	18.3	20.5	17.8	14.0
		MCDBR	18.7	19.1	22.3	23.9	22.9	21.8	22.1	21.9	22.1	22.6	21.9	18.7
		MCWBR	11.2	10.2	10.4	10.6	9.8	8.9	9.7	10.3	10.1	10.0	11.2	10.9
	5% WB	MCDBR	18.1	18.5	21.1	20.8	19.8	19.0	19.2	19.4	19.6	21.1	21.1	17.8
		MCWBR	10.9	9.8	9.9	9.4	9.1	8.4	9.9	10.1	9.9	9.7	10.9	10.6
Clear-Sky Solar Irradiance	taub	0.332	0.401	0.523	0.615	0.641	0.656	0.645	0.680	0.622	0.477	0.387	0.349	
	taud	2.132	1.946	1.654	1.506	1.507	1.525	1.566	1.502	1.568	1.790	1.984	2.072	
	Ebn at Noon	263	255	232	220	218	214	215	203	204	225	237	245	
	Edh at Noon	36	50	74	90	92	90	86	90	80	58	42	36	
All-Sky Solar Radiation	RadAvg	898	1175	1624	1920	2141	2133	2003	1823	1674	1464	1077	862	
	RadStd	111	86	99	73	102	105	156	158	94	41	55	56	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+1.03	N/A	N/A	N/A	-1.73	-2.30	N/A	-252	+272	+110
Regional (2 neighbors)		+0.68	N/A	N/A	N/A	-1.58	-1.44	N/A	-252	+272	+110

Nomenclature: See separate page